

AquaPulse — Real-Time Groundwater Resource Evaluation Platform

Complete Technical Documentation, Mathematical Engine, Architecture & API Reference

Project Name: AquaPulse	Team Name: Team Mythos
SIH Problem ID: SIH25068	Ministry: Ministry of Jal Shakti (CGWB)
Problem Title: Real-time Groundwater resource evaluation using DWLR data	Category: Software / IoT Telemetry Analytics
Live Website: https://sayanrooj.github.io/mythos/	GitHub Code: https://github.com/sayanrooj/mythos

1. Problem Statement & Background Context

Under the National Hydrology Project (NHP), India's Central Ground Water Board (CGWB) maintains ~15,000+ **telemetric Digital Water Level Recorder (DWLR) piezometers** logging hourly water level depths across major aquifers (Alluvial, Hard-Rock, Coastal). While readings stream to India-WRIS, traditional assessments are conducted retrospectively (annual/periodic), causing acute agricultural extraction surges and unseasonal depletion to go unaddressed for months. Furthermore, field sensors suffer from communication dropouts and electronic noise. **AquaPulse** provides an automated, real-time evaluation and early-warning engine.

2. System Architecture (5 Decoupled Layers)

Layer	Technology	Function & Operational Role
1. Ingestion Layer	Strategy Adapter (Python)	Decoupled ingestion supporting simulated SQLite (default) and live India-WRIS REST API feeds.
2. Analytical Engine	NumPy, SciPy, Math	2-metric stress classification, 4 anomaly detectors, and Holt Linear 30-day exponential forecasting.
3. REST API Gateway	Flask 3.1, CORS, SQLite	High-speed microservice providing /stations, /dashboard/summary, /forecast, and /alerts/dispatch.
4. Frontend UI	React 18, Vite, Leaflet, Recharts	National Overview Dashboard, interactive pulsing geospatial map, detail drawers, and educational guide.
5. Emergency Gateway	SMS & NIC Email Gateways	Automated multi-channel dispatch routing emergency alerts to District Nodal Hydrologists & DDMAs.

3. Scientific Stress Classification & Anomaly Logic

Metric A (Deviation): Deviation = Current Level (m bgl) - 12-Month Baseline Mean (m bgl).

Metric B (Decline Rate): 30-day linear regression slope (m/day) computed via Ordinary Least Squares (OLS).

Final Stress Level: MAX severity of Metric A and Metric B.

Category	Metric A Threshold	Metric B Threshold	Action Triggered
■ Safe	≤ 2.0 m	≤ 0.010 m/day (≤ 30 cm/mo)	Normal water balance; routine monitoring.

■ Semi-Critical	2.0 – 5.0 m	0.010 – 0.030 m/day	Crop water advisory & weekly telemetry review.
■ Critical	5.0 – 10.0 m	0.030 – 0.060 m/day	Block-level extraction audit & warning alert.
■ Over-Exploited	> 10.0 m	> 0.060 m/day (> 1.8 m/mo)	Emergency multi-channel SMS/Email alert to DDMA.

4. Holt Linear Forecasting & Anomaly Detection

Holt's Linear Exponential Smoothing (30-Day Projections): Level equation $L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + T_{t-1})$, Trend equation $T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$, Forecast $\hat{Y}_{t+h} = L_t + hT_t$ with 95% confidence bands $\pm 1.96 \cdot \sigma \cdot \sqrt{h}$ (Hyperparameters: $\alpha=0.3$, $\beta=0.1$).

Anomaly Detectors:

- **Gaps:** >6 consecutive missing readings (telemetry dropout) → WARNING/CRITICAL alert.
- **Spikes:** Single-reading jump >5.0m (electronic fault) → WARNING alert.
- **Sustained Decline:** 7-day slope >0.05 m/day (over-extraction event) → CRITICAL alert.
- **Status Downgrade:** Transition from Safe into stressed tiers → Automated alert dispatch.

5. API Endpoints & Codebase Structure

Endpoint	Method	Description & Output
/stations	GET	Returns all 30 DWLR stations with coordinates, aquifer type, latest level, and stress status.
/dashboard/summary	GET	State-wise stress distribution, 12-month national daily mean trend, and top-10 risk leaderboard.
/stations/:id/forecast	GET	30-day daily projected water levels with upper/lower 95% confidence bounds.
/alerts	GET	Active telemetry anomaly and status downgrade alerts filtered by type/severity.
/alerts/:id/dispatch	POST	Simulates automated multi-channel SMS & NIC Email dispatch to District Authorities.
/simulate/tick	POST	Ingests +1 hour of synthetic telemetry across all stations and re-evaluates anomalies in real time.
/stations/:id/export/csv	GET	Generates downloadable CSV file with 365 observed days + 30 forecast days.

6. Verification Test Results & India-WRIS Live Integration

Pytest Verification: 33/33 Unit & Integration Tests Passed (100% pass rate in 0.11s).

Production India-WRIS Switch: The data layer uses the Strategy Pattern in backend/data_source.py. Swapping simulated SQLite for live CGWB telemetry only requires setting DATA_SOURCE=india_wris with OAuth2 credentials.